sLight User Guide

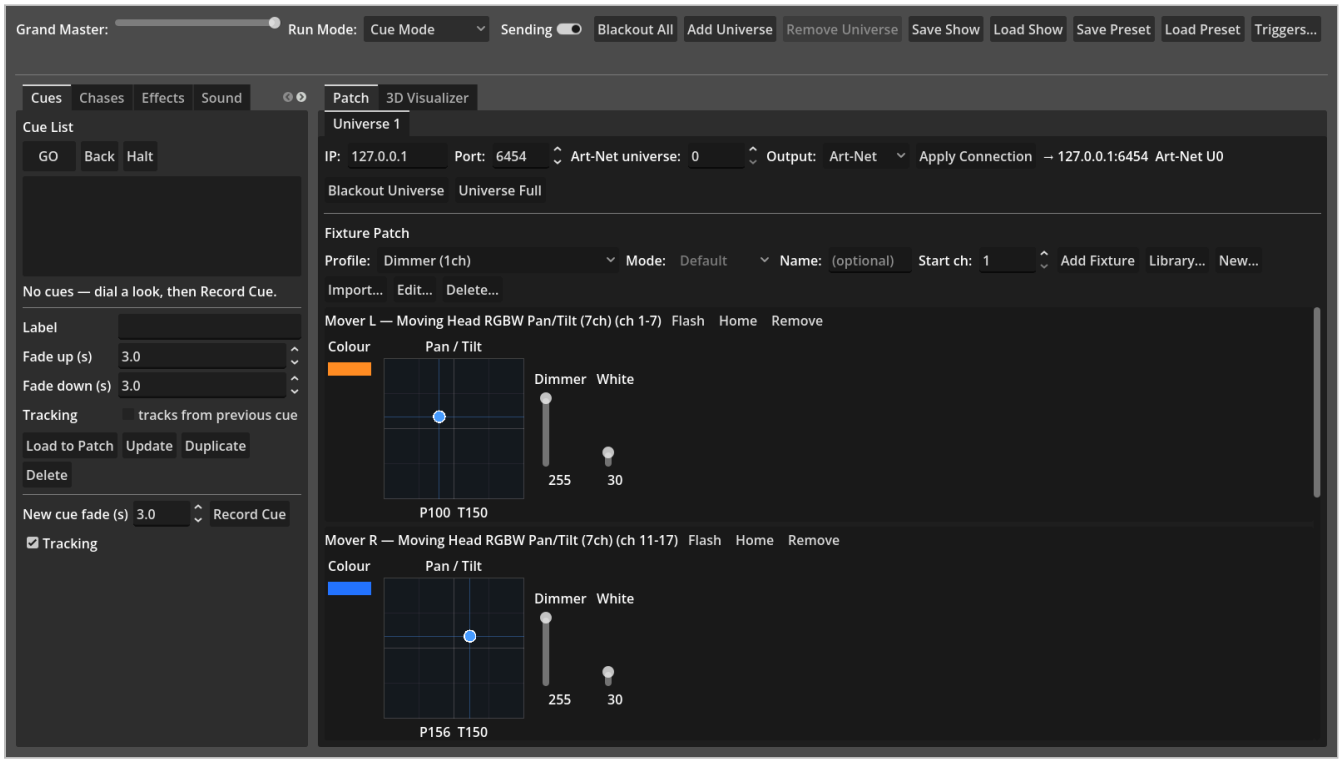
A step-by-step walkthrough for a new user, from opening the app for the first time to running a show. For a complete technical reference of every feature, see the README — this guide focuses on the essential workflow, in order, with screenshots.

What you need

sLight controls real (or simulated) DMX lighting fixtures. You don't need any hardware to follow this guide — see Testing without real hardware in the README for a free way to watch the DMX output live (QLC+ or an Art-Net monitor). When you're ready to control real lights, see Going to real fixtures.

Launch the sLight application — there's no separate setup step, no account, and no configuration file to edit first. The whole window you'll see below is what opens.

The window, at a glance



The window is split into three areas:

- **Top bar** — the **View** selector (**Console**, **DJ Edit Mode**, **DJ Perform Mode**), Grand Master fader, the **Run Mode** selector, **Sending** toggle, Add/Remove Universe, New/Save/Load Show and Preset, and **Triggers…** for MIDI/OSC. These are always visible, whatever tab or view you're on.
- **Left side — playback tabs**: **Cues**, **Chases**, **Effects**, **Sound**, **Groups**. This is where you build and run a show.
- **Right side — Patch / 3D Visualizer**: one tab per universe for patching fixtures and dialing them in by hand, and a 3D view of your rig lit live by whatever the console is currently outputting.

That split is the **Console** view. The top bar's **View** selector swaps it for **DJ Edit Mode** — where you build a light show for each song in your music library — or **DJ Perform Mode**, where VirtualDJ plays those shows live (Steps 10 and 11). The console keeps running underneath either one.

Everything below walks through these in the order you'd normally use them for the first time.

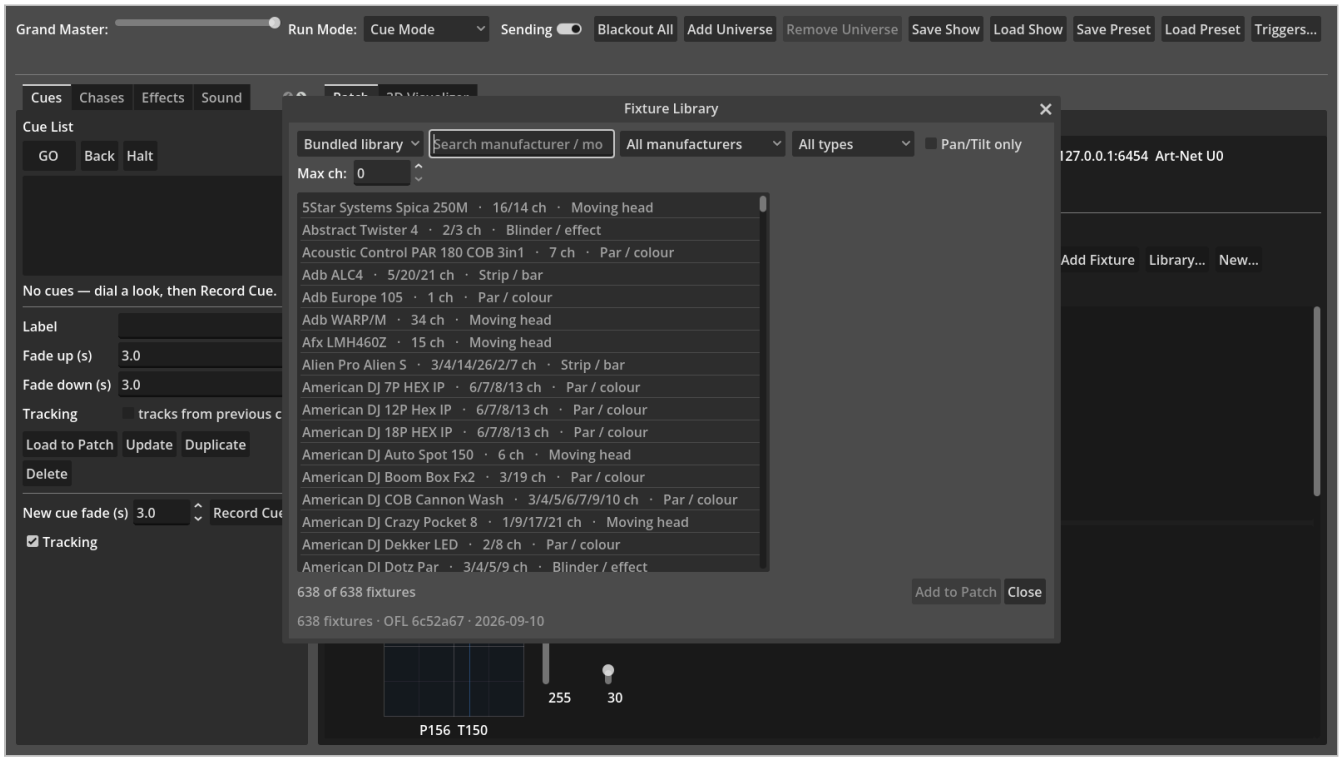
Step 1 — Add a universe and connect it

Each **Universe** tab (right side) is one 512-channel DMX universe. A fresh project starts with **Universe 1**. Pick an **Output** — Art-Net is the default and needs no extra hardware or plugins — set the target (for Art-Net, the IP/Port/universe number; `127.0.0.1:6454` universe `0` talks to a monitor running on the same machine), and click **Apply Connection**. Add more universes from **Add Universe** in the top bar (or **Ctrl+Shift+N**) if your rig needs more than 512 channels; **Remove Universe** (**Ctrl+Shift+R**) asks first, telling you how many fixtures go with it.

Step 2 — Patch your first fixture

"Patching" tells sLight what kind of light is plugged in where. In the **Fixture Patch** row, pick a **Profile** (a description of the fixture's DMX channels) and, if it has more than one, a **Mode**. Set the **Start ch:** to where it sits in the universe, optionally give it a **Name**, and click **Add Fixture**. Its control panel appears right below — a colour picker, a pan/tilt pad for moving heads, sliders for everything else (see the screenshot above).

A handful of generic profiles (Dimmer, RGB, RGBW, a moving head) ship built in, but real fixtures vary a lot — click **Library…** to search ~640 real fixtures converted from the Open Fixture Library:



Search by manufacturer or model, filter by type/channel count/pan-tilt, preview a fixture's modes and channel layout, then **Add to Patch**. If you have a GDTF or Open Fixture Library file for something not in the bundle, **Import…** reads it directly.

Step 3 — Control a fixture by hand

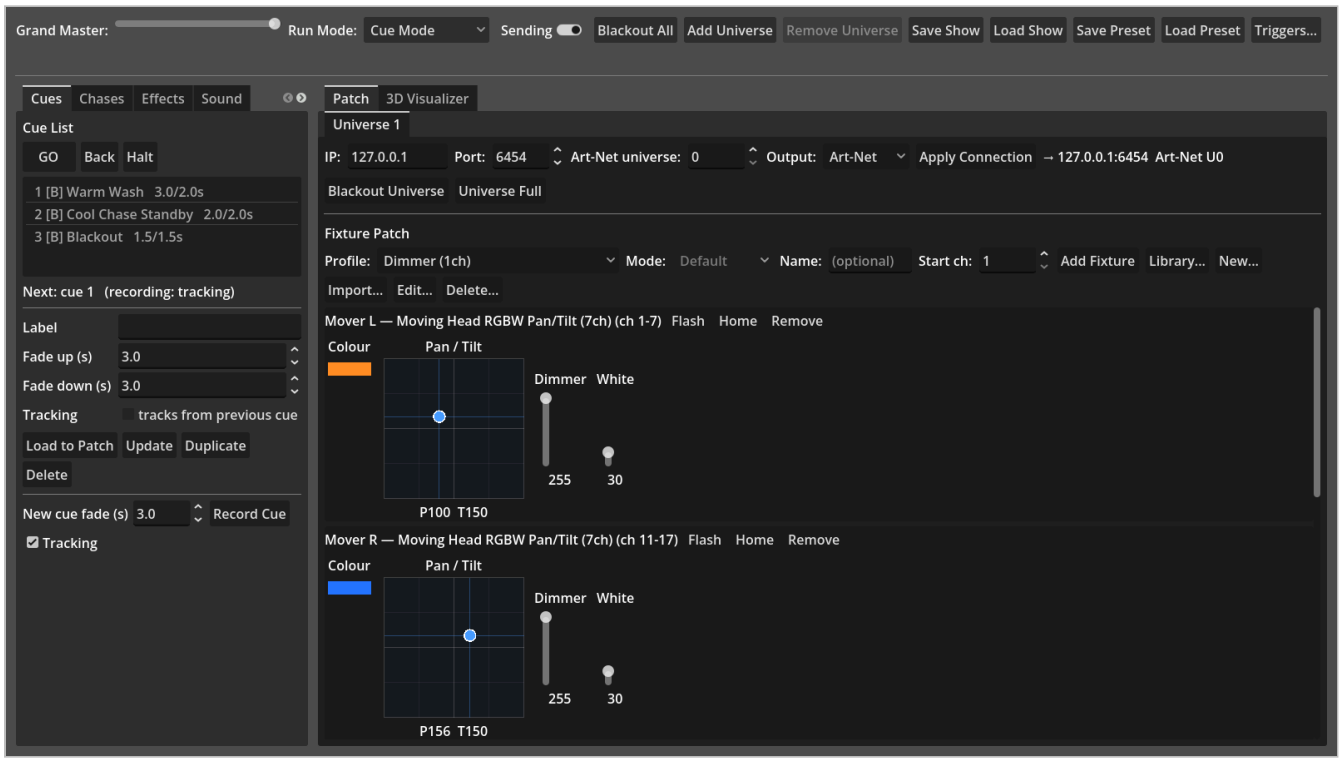
Every patched fixture gets purpose-built controls (visible in the screenshot for Step 2):

- A full Red/Green/Blue trio becomes one **colour picker**.
- A **Pan** + **Tilt** pair becomes a 2D pad — drag the puck to aim the head.
- Everything else gets its own slider, clamped to its usable range.

Flash (hold) snaps that fixture to full white and restores it exactly on release — handy for finding a fixture in a dark room. **Home** resets every control to its default. Fixture controls send live DMX output immediately (roughly 30 times a second, while **Sending** is on) — there's no separate "apply" step.

Step 4 — Record your first cues

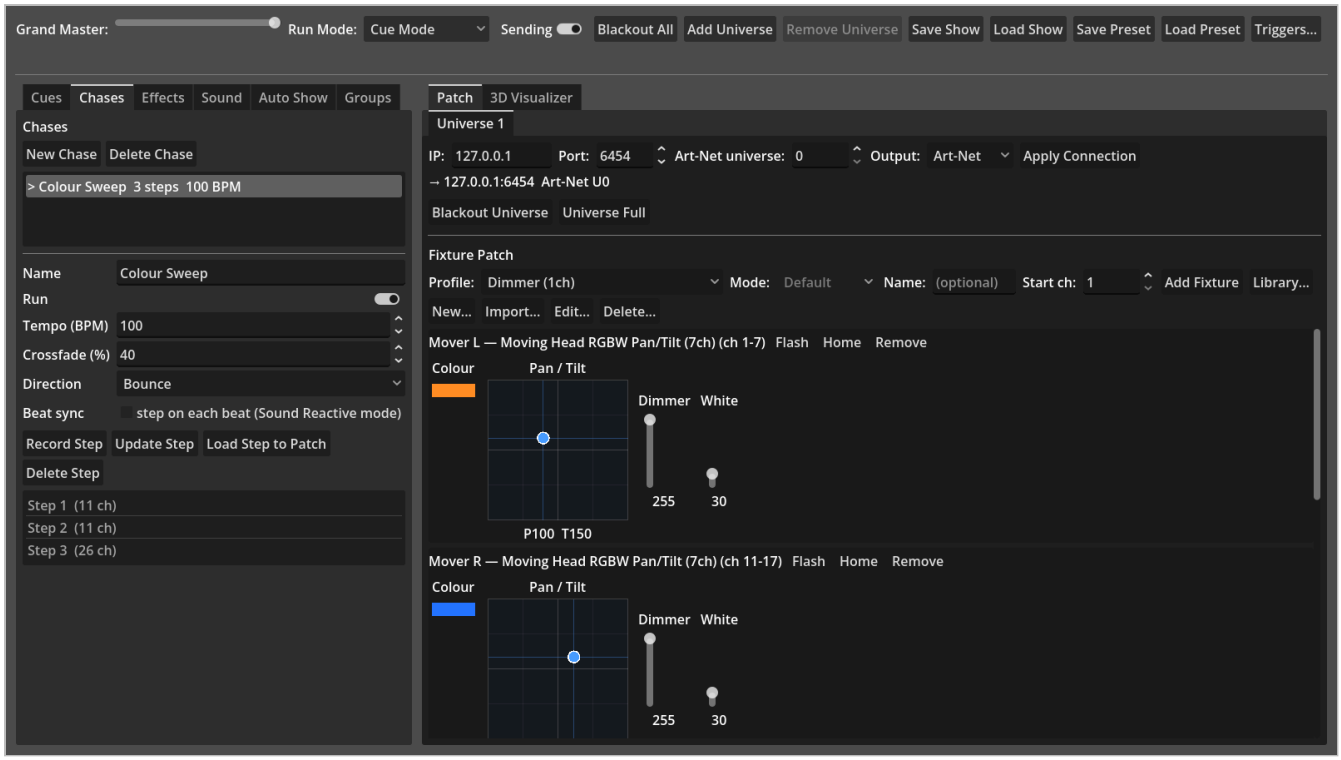
Once you've dialed in a look you like, switch to the **Cues** tab and click **Record Cue** — it captures the current output of every universe as a new cue. Repeat for a few different looks, then step through them with **GO** (or just press the spacebar) — the whole rig crossfades to the next cue's look over its fade time.



Select a cue to edit its **label** and **fade up/down** times inline. **Load to Patch** pulls a cue's stored look back into the fixture controls so you can tweak it and **Update** to re-record. See the README's Cue list section for the tracking vs. block distinction, which matters once you're editing a list of cues that build on each other.

Step 5 — Add movement with a chase

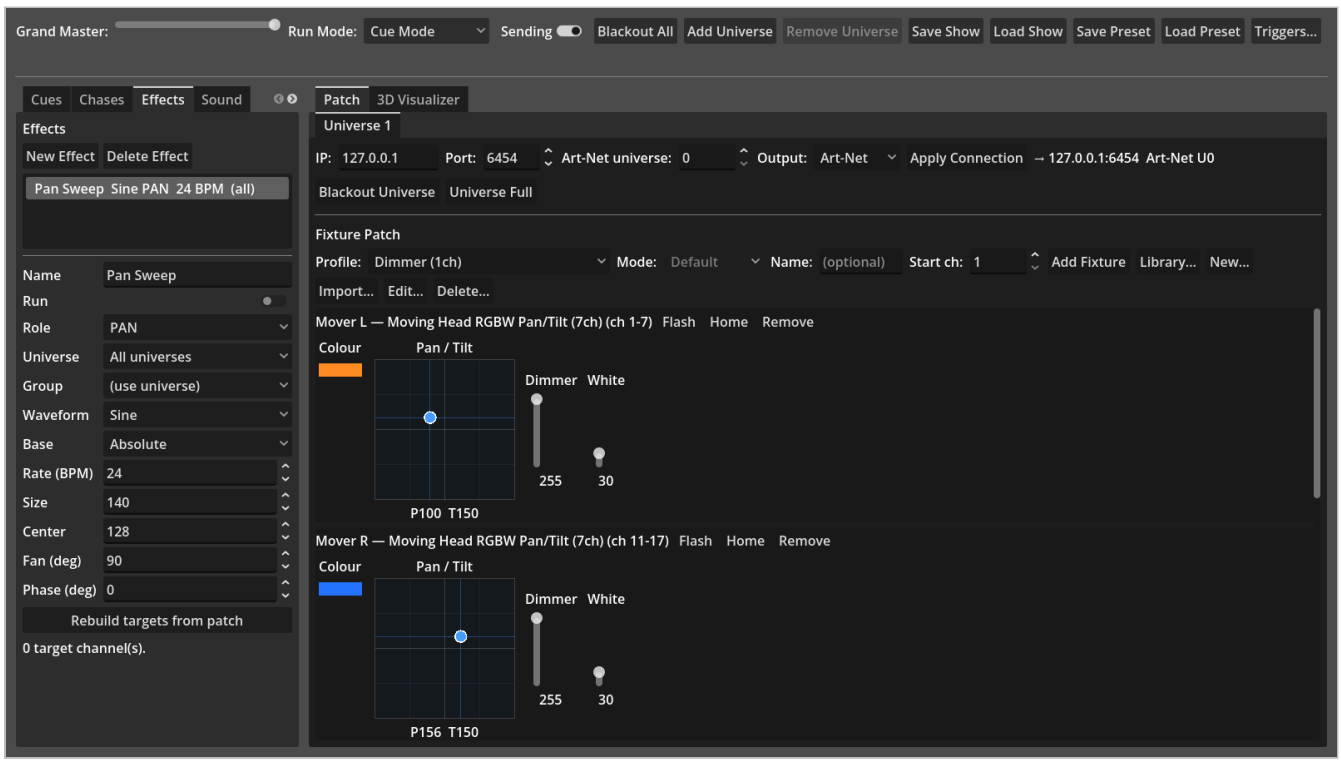
A **chase** is a list of captured steps that cycle automatically at a tempo — good for a repeating colour or position pattern that would be tedious to step through by hand. On the **Chases** tab, click **New Chase**, dial a look, click **Record Step**, and repeat for each step in the pattern.



Set the **Tempo (BPM)**, how much each step **Crossfade**s in from the last one, and a **Direction** (forward / backward / bounce). Click **Run** to start it — a chase runs as a layer *on top of* whatever your cues are doing, so it doesn't disturb your programmed look, and stops cleanly when you turn it off.

Step 6 — Layer in an effect

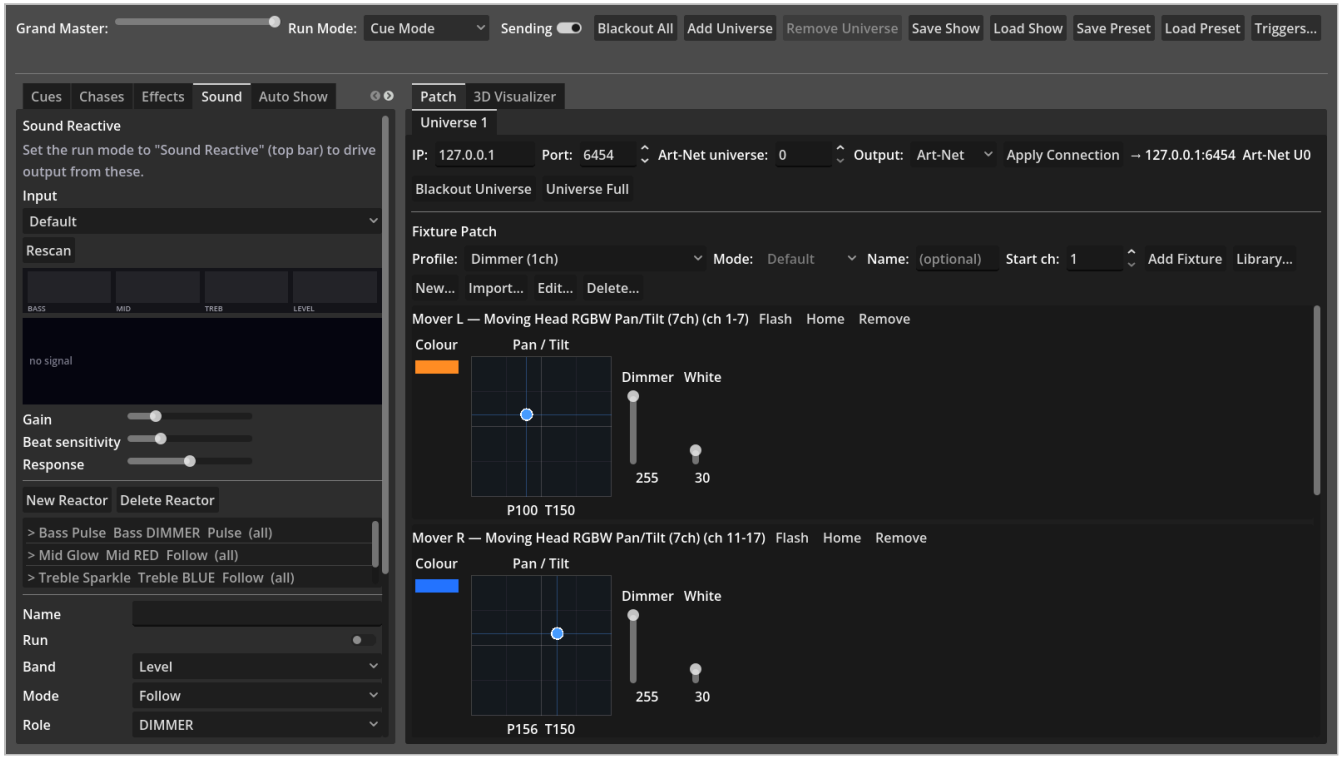
An **effect** is a smooth waveform (sine, triangle, sawtooth, square, or random) applied continuously to one channel role — a breathing dimmer, a Pan/Tilt circle, a slow colour swell — without needing to record any steps.



Pick the **Role** it drives and which fixtures (a **Universe**, or a **Group** — see Step 8), a **Waveform**, **Rate (BPM)**, and **Size** (how far it swings). **Fan (deg)** spreads the phase across the fixture list, so instead of every fixture moving in lockstep you get a chase-like ripple across the rig — try it with two Pan/Tilt effects 90° out of phase for a circle. Click **Run** to start it; like chases, effects layer over your cues without disturbing them.

Step 7 — React to music live

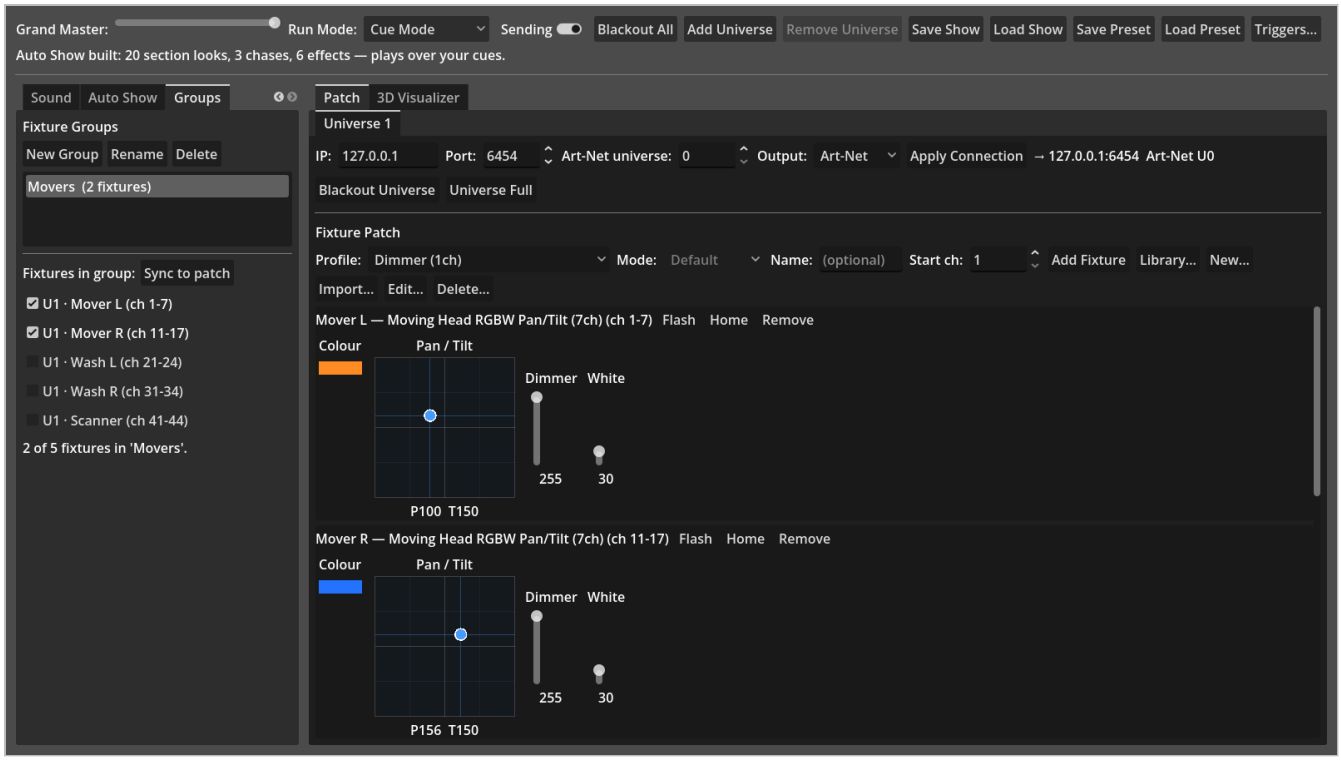
The top bar's **Run Mode** has two settings: **Cue Mode** (the default — the cue list drives playback) and **Sound Reactive**. Set it to **Sound Reactive** and switch to the **Sound** tab to drive output straight from a microphone or line input.



Pick an **Input** device, and a live **bass / mid / treble / level** meter with a spectrogram shows what's coming in. A **reactor** maps one band (or the beat) onto a channel role — **Follow** mode tracks the band continuously (bass into a dimmer swell), **Pulse** mode snaps to full on every detected beat and releases (a beat-synced flash). A starter set of four reactors is already here to try — **Bass Pulse**, **Mid Glow**, **Treble Sparkle**, and **Beat Strobe** (off by default, since it's a lot more intense than the others) — edit or delete them, or add your own with **New Reactor**. Tick **Run** on the ones you want active.

Step 8 — Organize fixtures into groups

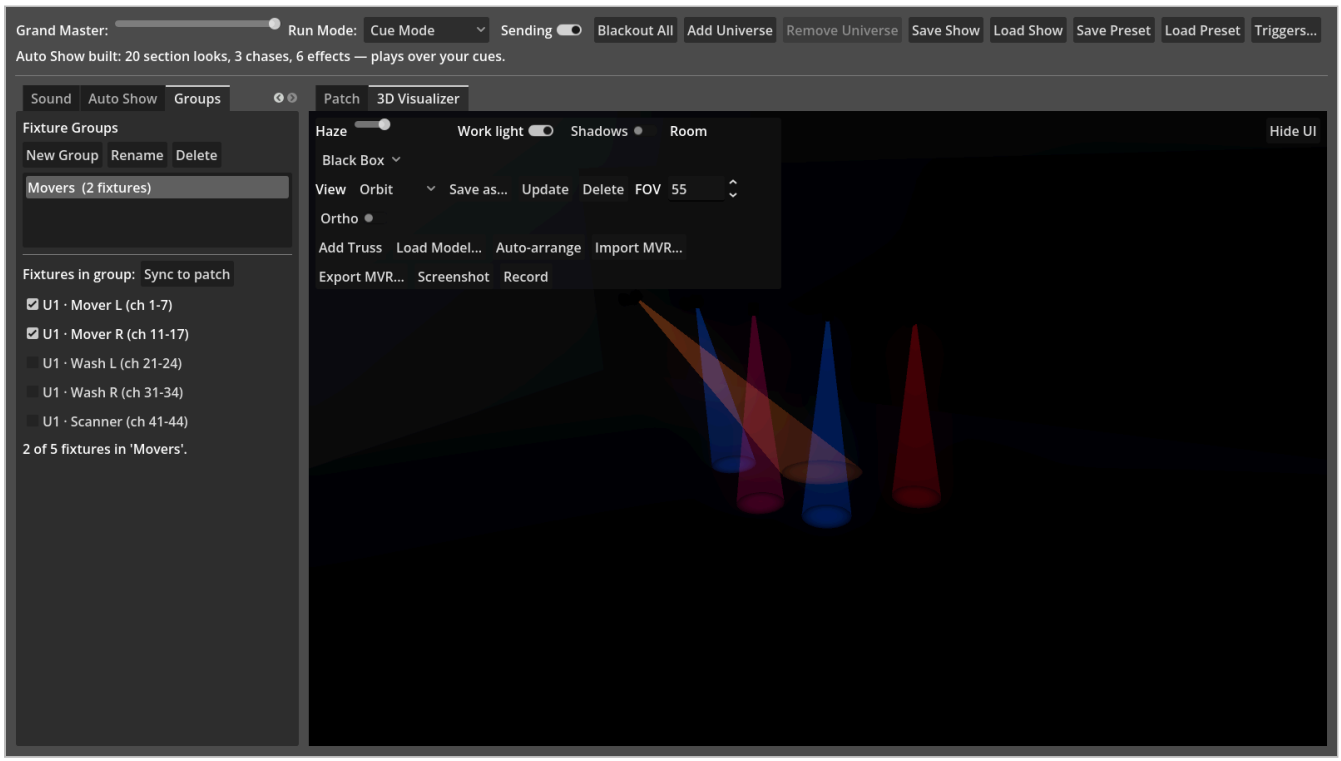
The **Groups** tab holds named sets of patched fixtures, so an effect or reactor can target just "the movers" or "the front truss" instead of a whole universe.



New Group, then tick which patched fixtures belong to it. Pick that group's name in an effect's or reactor's **Group** field instead of a universe filter. **Sync to patch** refreshes the checklist after you add or remove fixtures.

Step 9 — See it in 3D

The **3D Visualizer** tab (right side, next to Patch) shows your rig lit live by whatever the console is currently outputting — cues, chases, effects, and the grand master all show, updating continuously.

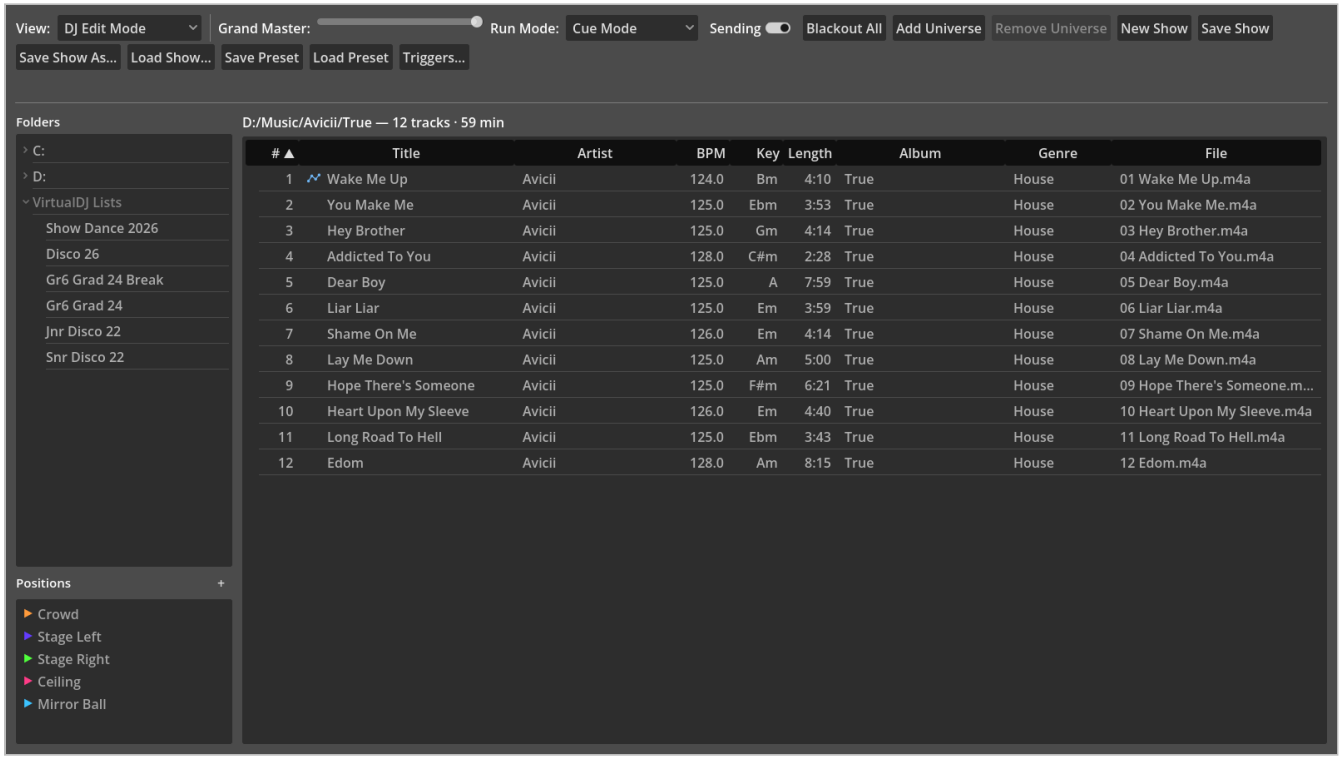


Left-drag empty space to orbit the camera, wheel to zoom. **Auto-arrange** hangs every patched fixture in a tidy grid to start from; drag a fixture on the floor (or type exact X/Y/Z and heading/tilt) to place it precisely. Got it wrong? **Undo** and **Redo** (or Ctrl+Z / Ctrl+Y with the pointer over the visualizer) step back and forth through your moves — a whole drag, each typed value, or an Auto-arrange is one step each. Any fixture with pan or tilt channels hangs in a moving yoke and follows them, whatever its category. **Add Truss** and **Load Model…** bring in rigging and set pieces. **Haze** adds real volumetric fog so beams show up in the air; **Room** switches between a few preset spaces. **Screenshot** and **Record** save a still or an MP4 straight from this view. Drag the **3D Visualizer** tab off the tab bar to pop it into its own window — handy for a second monitor at front-of-house.

Step 10 — Build a light show for a song (DJ Edit Mode)

Switch the **View** to **DJ Edit Mode** to program lighting against individual songs rather than cue by cue.

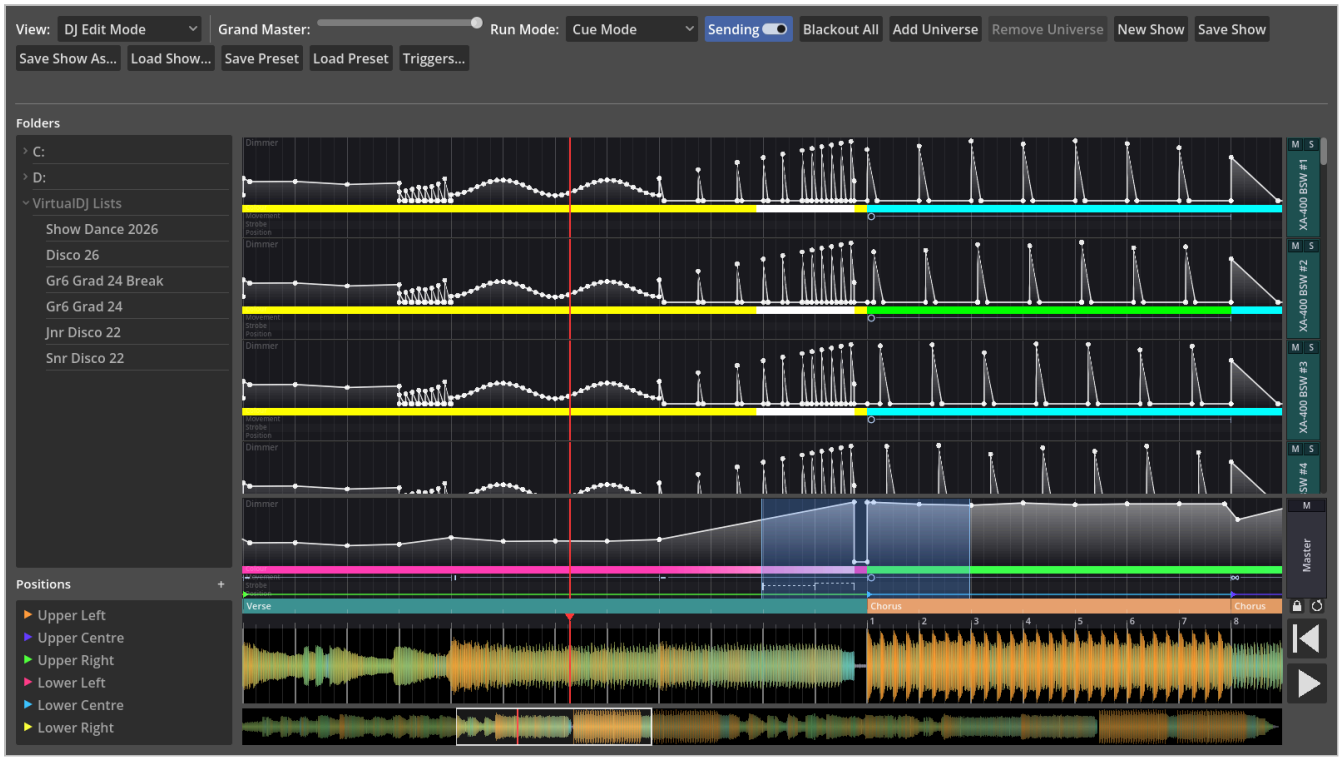
Find a song. On the left is a tree of this computer's drives and folders, followed by **VirtualDJ Lists** (your VirtualDJ Virtual Folders). Select one and its music is listed on the right — title, artist, BPM, key, length, album, genre — with details from VirtualDJ's database where it knows the track. Click a column title to sort. Songs that already have a light show in the current show are marked with a small blue node-line icon before the title. M4A / AAC songs open on Windows and macOS.



Open its timeline. Double-click a song. The list is replaced by its timeline (select a folder to go back). It opens fully zoomed out:

- Along the bottom is a colour waveform of the song, with VirtualDJ's beat grid drawn behind it and the bars numbered along the top. The mouse wheel zooms, **Shift**+wheel scrolls, and dragging with the middle mouse button pans. The slim overview beneath shows the whole song — drag its rectangle, or click to jump.
- **Play / Pause** (or **Space**) and **Back to start** sit to the right. The red playhead runs down every timeline; click or drag it on the waveform, the bar ruler or any timeline to move it.

- Above the waveform is the song's **structure** — Intro, Verse, Chorus and so on — worked out in the background when the song opens. It's locked by default (and locks again whenever you open another song); click the padlock to fix a section border the analyser put slightly wrong (drag it), or right-click a section to relabel, split or delete it. Right-click > **Snapping** chooses whether borders and selections snap to bars, beats or nothing.
- Above the structure is the **master** timeline, which drives every fixture, and above that a scrolling timeline for each patched fixture (and each head of a multi-head fixture).



While the song plays, the lights and the 3D Visualizer follow the playhead exactly as they will in a live show — edits included. When it's paused, the look at the playhead holds, so you can scrub through and see each moment.

Let sLight start it for you. Click **Generate** (beside the padlock) to build a complete show from the song's structure and energy. The master's dimmer follows the music section by section, rising and falling with its loudness bar by bar, and every 8 bars the look changes: the fixtures breathe, take turns, hit on the bar or chase across the rig on the beat as the energy rises, often with a fill in the last bar. Their hits follow the bass, and bright, trebly passages sparkle. Where the music drops out within a section the lights calm down with it, and kick back in when it returns. Each kind of section has its own colours, laid across the fixtures in splits, gradients, sweeps or chases. Before a big rise in energy the lights build: the master climbs to full, the chases speed up, the colours fade towards white and a strobe runs, sometimes ending on a beat of blackout before the drop. Breakdowns go dark and cool, with one or two fixtures at a time or slow swells. Moving heads move to your named positions by energy — the ones that spread the beams widest for the loudest parts — changing every 8 bars when it's loud, with movement that grows with the energy, the heads sometimes fanned out across it. Sections that come back build each time, and the last time the loudest one comes round is the finale, with everything. Not keen on the result? Click **Generate** again for another variation. Each press replaces what the timelines had, as a single undo step — a good starting point to edit by hand.

Generate just part of it. With areas selected on the timelines (see *Select, then edit* below), **Generate** only fills those areas, row by row, and leaves the rest of the song as it was — handy for redoing one chorus

without losing the edits around it. **Shift**-click **Generate** to analyse the song's structure afresh first (dropping any changes you made to it), then generate from the new structure; both happen as one undo step, and if the analysis fails the old structure is kept.

Generate casts each fixture by its category (chosen in the Patch tab). Washes carry the bulk of the colour. Movers carry the movement, usually in the palette's accent colour. Multi-cells play chases, waves, bar graphs and colour patterns across their heads. A mover with several heads moves as one, but now and then its heads break out into multi-cell chases and colours while it moves. Primary fixtures drive the show, and secondary ones support them with a calmer pattern in the palette the other way round, a little dimmer, their movers mirroring the primaries, following slower or holding still. Before a drop, in the finale and in some of the loudest moments, primary and secondary play as one. The **1–4** category group buttons split a category up: fixtures in the same group often step together, so a chase or a fan can run group by group (left and right, front and back). To keep a fixture out of the generated show altogether, toggle on the struck-through wand button beside its category. It stays in the patch, but Generate clears its timelines (and its heads') instead of filling them. It ignores the master timeline too, so it plays nothing from the generated show. Anything you then draw on its own timelines still plays.

Each timeline has rows for what its fixture can do:

- **Dimmer** — a line of nodes. Drag a node to move it (hold **Shift** to lock it to one axis), drag the line to raise or lower a stretch, **Ctrl**+click the line to add a node and **Ctrl**+right-click a node to remove it. The master's line starts at 0%, so nothing lights until you raise it. A fixture's (or head's) dimmer row starts empty and follows the master; give it values and it overrides the master over that stretch.
- **Colour** — segments of one static colour or a two-colour gradient. A fixture whose colour comes from a colour wheel turns its wheel to the nearest colour slot, and **Generate** gives it only its wheel's colours, changing from one to the next.
- **Strobe** — stretches where the strobe runs at Slow, Medium or Fast.
- **Position** (moving heads) — which named position the heads point at (see *Named positions* below).
- **Movement** (moving heads) — a Circle, Figure 8, Pan Sweep or Tilt Sweep around that position, Small / Medium / Large, cycling every 1, 2 or 4 bars — or None to hold still. They're kept small and slow enough for real moving heads to follow.

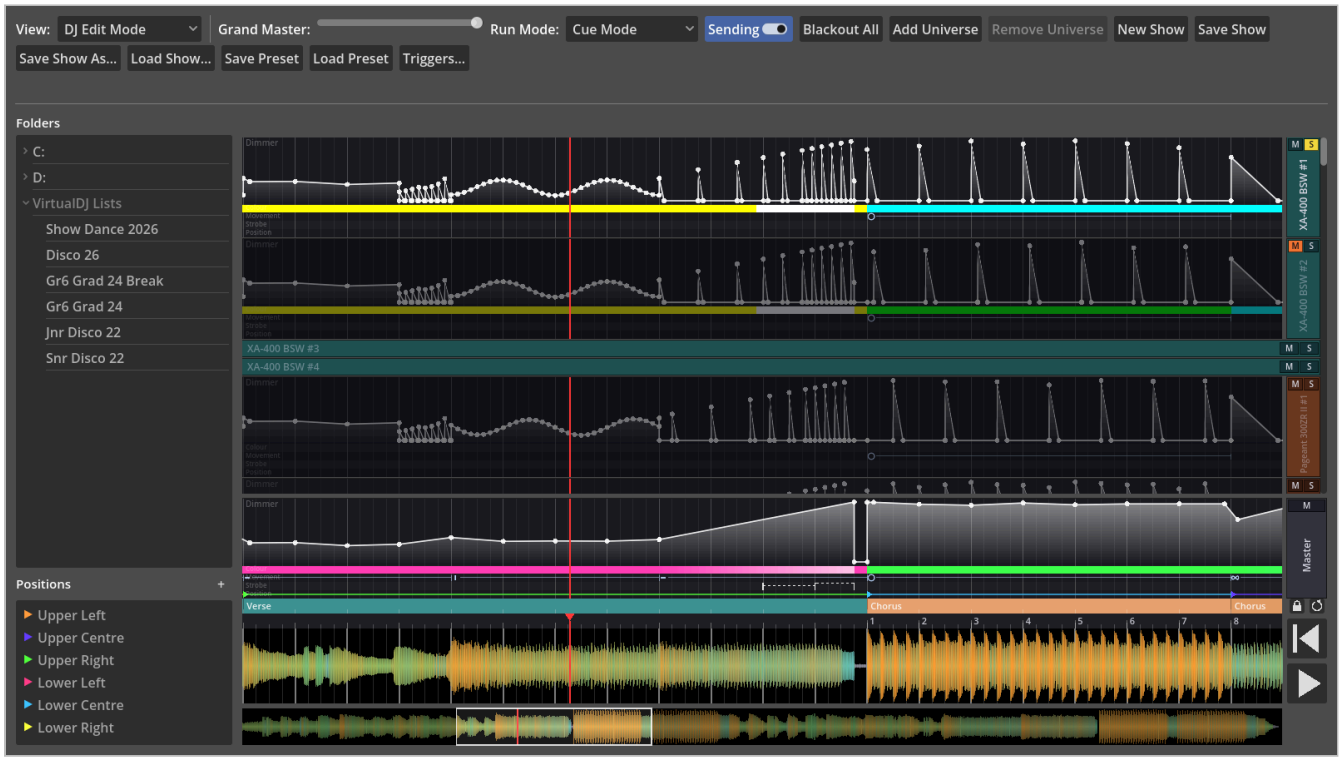
Select, then edit. Drag across empty parts of the timelines to select an area — over every row the drag crosses, snapped to bars or beats. **Ctrl**-drag adds another area, **Shift**-drag extends the last one. **Right-click** the selection for a menu of what fits: Dimmer shapes (Black, Half, Full, Fade In / Out, Pulse In / Out, Smooth, Pulse on Every Beat / Bar), Colour (Static Colour…, Gradient…, Rainbow, presets), Strobe speeds, Position, Movement (with Size and Cycle), and Clear for each. Drag a selection's start or end to resize it (what sits on that edge moves with it), or drag its inside to move its content elsewhere — a ghost shows where it will land.

Keys	Does
Delete	Clear the selected areas
Ctrl+C / Ctrl+X / Ctrl+V	Copy / cut / paste — into every selected fixture at once, or at the playhead with nothing selected
Ctrl+Z / Ctrl+Y	Undo / redo (each song keeps its own history, up to 100 steps)
Ctrl+A	Select everything
Escape or **Ctrl+D**	Clear the selection

Mute, solo and tidy up. Each timeline's name plate, at the right, has an **M** (mute) button, and every one but the master's an **S** (solo).

- **Mute** leaves a timeline out of playback. Muting the master stops everything; muting a fixture mutes its heads too; muting a head silences just that head.
- **Solo**: while any timeline is soloed, only the soloed ones play. A soloed head still gets its fixture's and the master's timelines, since it builds on them.
- Timelines left out are drawn faded, and their channels are left to the console, as if the song didn't cover them.

Click a fixture's or head's name plate (away from M / S) to **collapse** its timeline to a slim bar with its name — collapsing a fixture folds its heads away with it, though each head can then be expanded on its own. Click the bar to expand it again. A collapsed timeline still plays, but can't be edited or selected, so selections pass it by. It's a quick way to get the fixtures you aren't working on out of the way.



Mute, solo and collapse are saved with each song's timelines (so every song keeps its own layout) and DJ Perform Mode plays the song the same way, but they aren't undo steps — Ctrl+Z leaves them as they are.

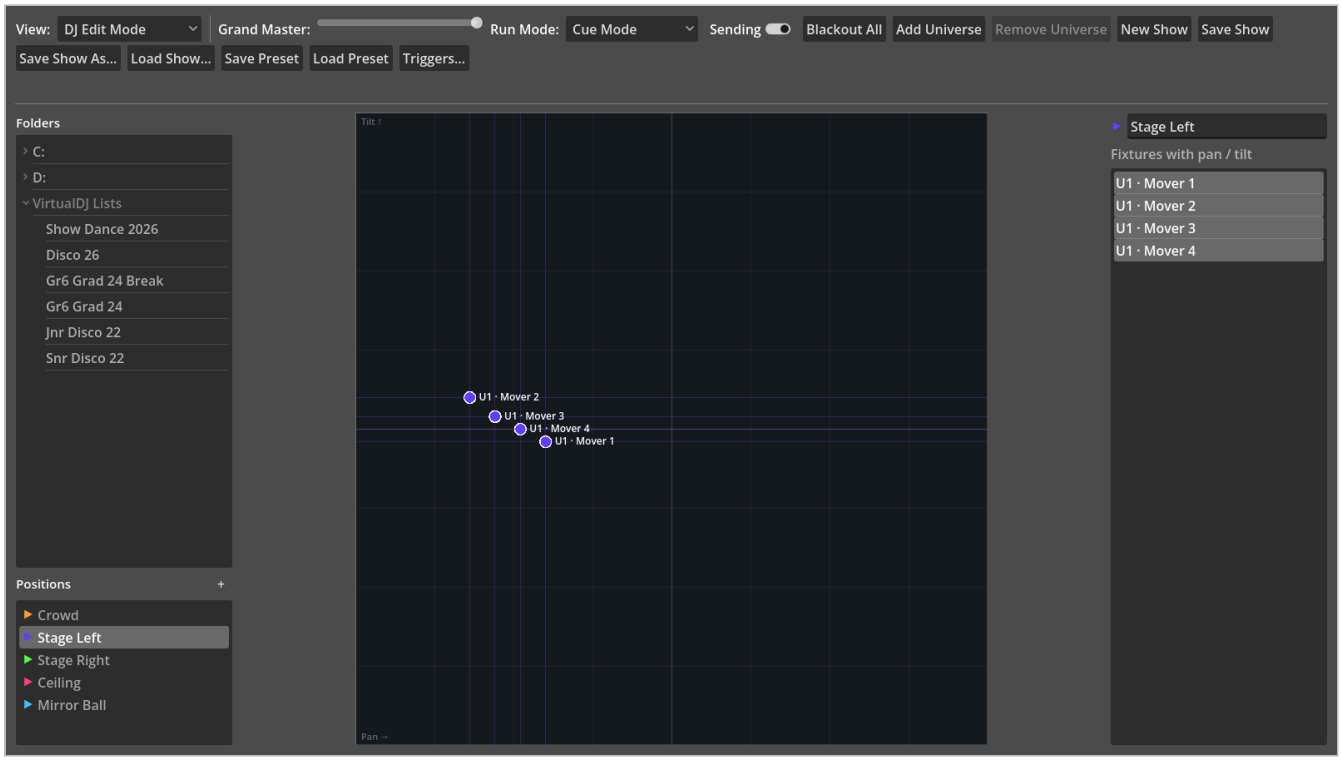
A song's timelines are saved automatically when you leave it — open another song, switch view, or close the app — and with **Save Show**. They're stored with the show (see *Saving your work*), keyed so a song

that's been moved or renamed keeps its light show.

Named positions

Moving heads point at **named positions** rather than raw pan/tilt values. In DJ Edit Mode's left-hand tree, below the folders, the **Positions** header lists them; click its **+** to add one. It opens in the **position editor**: a large pan/tilt pad beside a list of every moving fixture. Select fixtures in the list (**Ctrl** / **Shift** for several) and their points light up on the pad in the position's colour — drag one to aim them all. While selected, those fixtures are lit full white and pointed live at the position, so you can see exactly where they land in the room; they go back to how they were when you leave. Rename the position at the top of the editor, and right-click one in the list to **Rename**, **Duplicate** or **Delete** it.

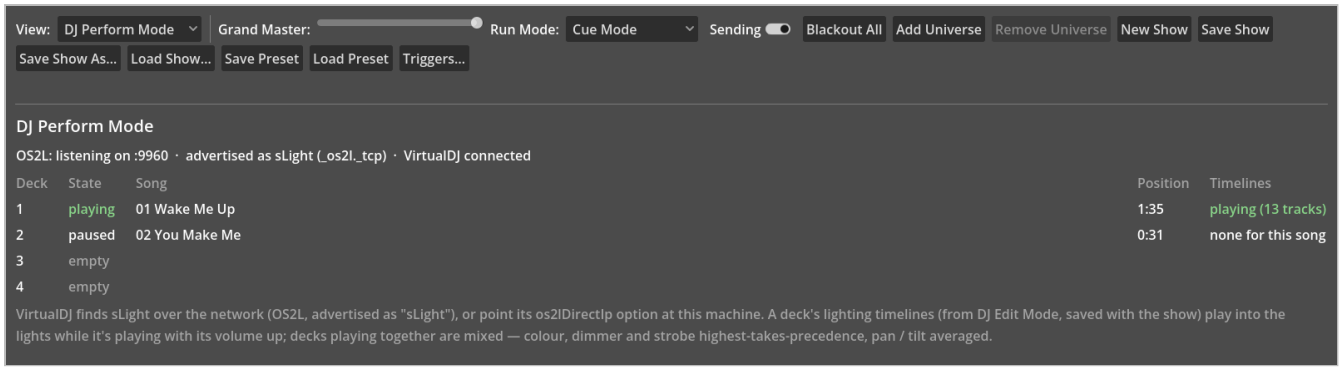
To reuse aim points, select fixtures and press **Ctrl+C** to copy their points, then select fixtures — in the same position or another one — and press **Ctrl+V** to paste. The selected fixtures take the copied points in list order, one each, starting again from the first if you've selected more fixtures than you copied; so copying one fixture and pasting onto several aims them all at the same spot. (While you're typing in the name field, Ctrl+C / Ctrl+V copy and paste its text instead.)



Use them in a timeline's Position row (right-click > **Position**), or let **Generate** spread them across the song: it saves the positions that put the beams furthest apart for the loudest parts.

Step 11 — Play it live with VirtualDJ (DJ Perform Mode)

Switch the **View** to **DJ Perform Mode** and sLight starts listening for VirtualDJ over OS2L. It's advertised on the network as "sLight" (port 9960), so VirtualDJ normally finds it on its own; otherwise point VirtualDJ's `os2lDirectIp` setting at this computer. The panel shows the connection and, for each of up to four decks, the song loaded, whether it's playing, its position and whether its timelines have loaded.

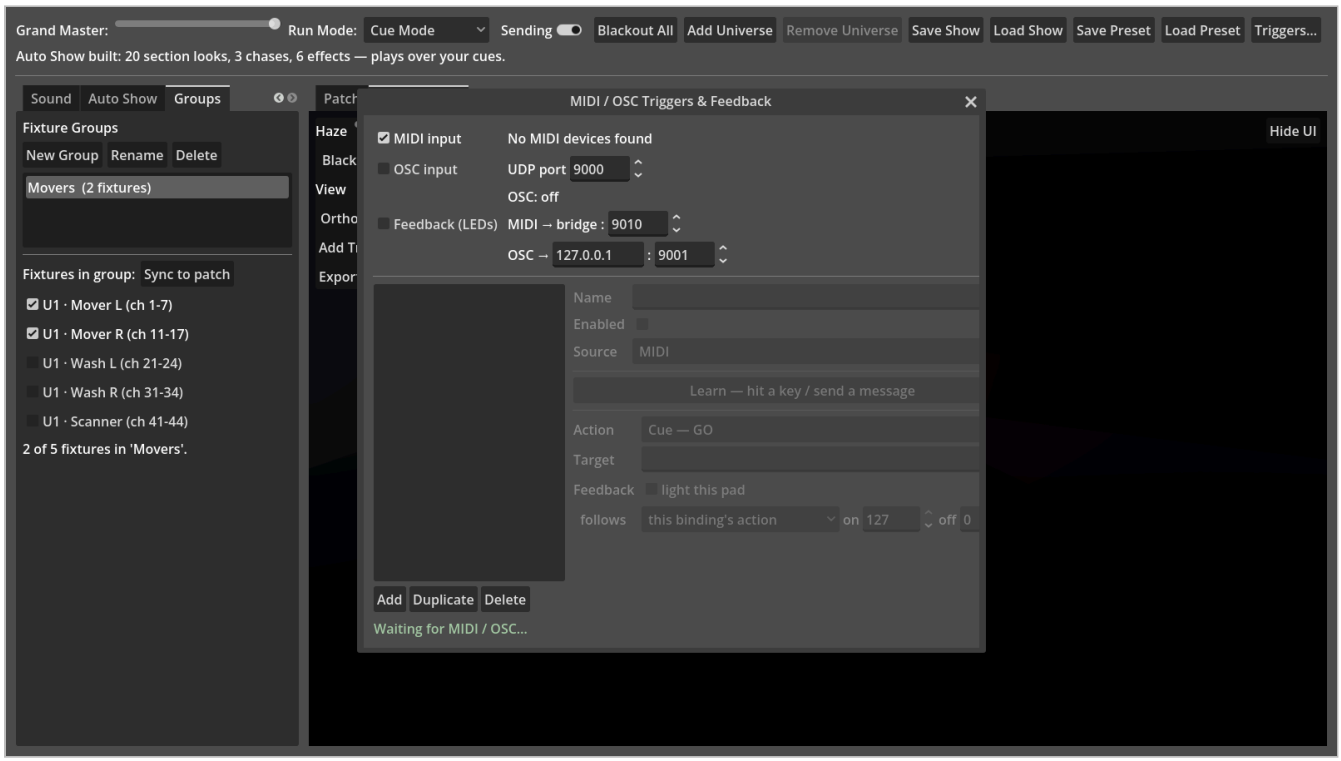


Whenever a deck loads a song, its light show from DJ Edit Mode loads in the background. A deck drives the lights while it's **playing with its volume fader up** — a track you're cueing in your headphones with the fader down stays dark — following VirtualDJ's position and pitch. When two decks play together during a mix, their shows blend: brightest wins for colour, dimmer and strobe, and pan/tilt are averaged.

Only what a song's timelines cover is driven; everything else stays as the console has it, and chases and effects still layer on top. Leaving DJ Perform Mode stops the listener and the playback.

Optional: MIDI and OSC control surfaces

Triggers… in the top bar (works from any tab) opens the binding editor, so a MIDI controller or an OSC app like TouchOSC can fire cues, toggle chases/effects, or trigger a blackout.



Turn on **MIDI input** or **OSC input**, click **Add**, then **Learn** and play the pad / move the fader / send the message you want bound — the match fields fill in automatically. Pick an **Action** (Cue GO, a chase or effect toggle, blackout, …). **Feedback (LEDs)** can light a pad back based on what it controls, or as a standalone beat/sending/run-mode indicator. This is entirely optional — most of the app works fine with just a mouse and keyboard.

MIDI feedback needs somewhere to send it: the screenshot above shows the default **MIDI → bridge :** field, which needs a small helper script running (`python tools/midi_bridge.py`, see the README). If the optional `native` extension is built, this row becomes a **MIDI output** port picker instead — pick a real MIDI port directly, no extra script needed.

Saving your work

A show lives in one `.slight` file you choose, holding everything — every universe's connection settings and patch, cues, chases, effects, groups, named positions, sound reactors, MIDI/OSC bindings and the visualizer's stage. The songs' lighting timelines go in a folder beside it (`MyShow.slight` → `MyShow.timelines/`), so keep the two together when you copy a show to another computer.

Top bar	Keys	Does
Save Show	**Ctrl+S**	Save to the current show file (asks where, for a new show)
Save Show As…	**Shift+Ctrl+S**	Save to a new file, copying the timelines folder alongside
Load Show…	**Ctrl+O**	Open a show
New Show	**Ctrl+N**	After asking, start an empty show (your output settings are kept)

The show you last saved or loaded reopens automatically the next time you start sLight. **Save/Load Preset** is a lighter-weight snapshot of just the live DMX output, for a quick "go back to how it looked a minute ago" without touching your cue list.

Keyboard shortcuts

Keys	Does
Space	GO the next cue (Console, Cue Mode) / play or pause the song (DJ Edit Mode)
Ctrl+N / Ctrl+O	New show / load a show
Ctrl+S / Shift+Ctrl+S	Save show / save show as
Ctrl+Shift+N / Ctrl+Shift+R	Add a universe / remove the current one

DJ Edit Mode's timeline editing shortcuts are listed in Step 10.

Where to go next

This guide covers the everyday path from an empty patch to a running show. The README is the complete reference — fixture profile details (multi-mode personalities, 16-bit fine channels, named value ranges for gobo/colour wheels), sACN and USB DMX output, cue tracking vs. blocks, MVR import/export, every run mode's exact behaviour, and exactly how DJ Perform Mode mixes decks and drives strobe, position and movement channels are all there if you want to go deeper.